

Data Types and Variables

In this lecture, we will explore the fundamental concepts of data types and variables in Python. Understanding these concepts is crucial for effective programming and data manipulation. By the end of this lecture, you will be able to identify different data types, create and use variables, perform arithmetic operations, and utilize built-in functions for data manipulation.

1. Introduction to Data Types

Data types are classifications that specify which type of value a variable can hold. In Python, the primary data types include: - **Integers**: Whole numbers, e.g., 5, -3, 42. - **Floating-point numbers**: Numbers with decimal points, e.g., 3.14, -0.001, 2.0. - **Strings**: Sequences of characters enclosed in quotes, e.g., 'hello', "Python Programming".

2. Working with Variables

A variable is a named piece of memory that can store a value. To create a variable, you use an assignment statement, which has the syntax:

```
name = value
```

For example: - `x = 5` - `gpa = 3.14`

Once a variable is assigned a value, you can use it in expressions. For instance: - If `x = 7`, then `x + 4` evaluates to 11.

3. Performing Arithmetic Operations

Python supports several arithmetic operations, which can be performed on integers and floating-point numbers. The basic arithmetic operators include: - **Addition (+)**: Adds two numbers. - **Subtraction (-)**: Subtracts one number from another. - **Multiplication (*)**: Multiplies two numbers. - **Division (/)**: Divides one number by another, resulting in a floating-point number. - **Integer Division (//)**: Divides and returns the integer part of the quotient. - **Modulus (%)**: Returns the remainder of a division. - **Exponentiation (**)**: Raises a number to the power of another.

Example of Arithmetic Operations:

```
>>> 3 + 7 # Addition
10
>>> 5 - 2 # Subtraction
3
>>> 4 * 2 # Multiplication
8
>>> 8 / 2 # Division
4.0
>>> 7 // 2 # Integer Division
3
>>> 7 % 2 # Modulus
1
>>> 2 ** 3 # Exponentiation
8
```

4. Understanding Data Type Conversion

Python allows for conversion between data types. You can convert an integer to a float and vice versa using the `int()` and `float()` functions: - `int(2.5)` results in 2. - `float(2)` results in 2.0.

5. Strings in Python

Strings are a vital data type in Python. They can be created using single or double quotes. You can

also use triple quotes for multi-line strings. Here are some important string operations: - **Concatenation**: Combining two strings using the + operator. - **Indexing**: Accessing individual characters using square brackets, e.g., s[0] gives the first character. - **Slicing**: Extracting a substring using the syntax s[start:end].

Example of String Operations:

```
>>> s = 'hello'
>>> s + ' world' # Concatenation
'hello world'
>>> s[1] # Indexing
'e'
>>> s[1:4] # Slicing
'ell'
```

6. Built-in Functions for Data Manipulation

Python provides several built-in functions that are useful for data manipulation: - len(): Returns the length of a string or list. - str(): Converts a value to a string. - int(): Converts a value to an integer. - float(): Converts a value to a floating-point number.

Example of Built-in Functions:

```
>>> len(s) # Length of the string
5
>>> str(10) # Convert integer to string
'10'
>>> int('5') # Convert string to integer
5
>>> float('3.14') # Convert string to float
3.14
```

7. Conclusion

In this lecture, we covered the essential data types in Python, how to create and use variables, perform arithmetic operations, and utilize built-in functions for data manipulation. Understanding these concepts is foundational for further programming in Python. As you practice, try to create your own variables, perform operations, and manipulate strings to solidify your understanding.

Quiz Questions

Q1: What are the primary data types in Python?

- Integers, Strings, Lists
- Integers, Floating-point numbers, Strings
- Strings, Tuples, Dictionaries
- Booleans, Sets, Arrays

Correct: Integers, Floating-point numbers, Strings

Q2: Which operator is used for integer division in Python?

- /
- //
- %
- **

Correct: //

Q3: What function would you use to find the length of a string in Python?

- size()
- length()
- len()
- count()

Correct: len()